Grade 5

FAST Mathematics

Sample Test Materials

The purpose of these sample test materials is to orient teachers and students to the types of paper-based FAST Mathematics questions. By using these materials, students will become familiar with the types of items and response formats they may see on a paper-based test. The sample items and answers are not intended to demonstrate the length of the actual test, nor should student responses be used as an indicator of student performance on the actual test. The sample test materials are not intended to guide classroom instruction.

Grade 5 FAST Mathematics Reference Sheet

DO NOT REMOVE THIS SHEET

Customary Conversions

1 foot = 12 inches
1 yard = 3 feet
1 mile = 5,280 feet
1 mile = 1,760 yards

1 cup = 8 fluid ounces
1 pint = 2 cups
1 quart = 2 pints
1 gallon = 4 quarts

1 pound = 16 ounces
1 ton = 2,000 pounds

Time Conversions

1 minute = 60 seconds
1 hour = 60 minutes
1 day = 24 hours
1 week = 7 days

Metric Conversions

1 centimeter = 10 millimeters
1 meter = 100 centimeters
1 meter = 1000 millimeters
1 kilometer = 1000 meters

1 liter = 1000 milliliters

1 gram = 1000 milligrams
1 kilogram = 1000 grams

Formulas

Rectangle	$P = l + l + w + w$ $P = 2l + 2w$ $A = l \times w$
Rectangular Prism	$V = l \times w \times h$ or $V = B \times h$

Key	
l = length w = width h = height B = area of the base	P = perimeter A = area V = volume

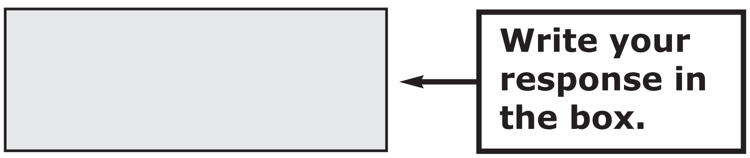
Use the space in this Test and Response Book to do your work. Then, completely fill in the bubble beside the answer you choose. For some items, filling in more than one bubble may be required, so read each item carefully. If you change your answer, be sure to erase completely.

Some items will ask you to write a response in a shaded box or boxes. See the sample item below.

Sample Item:

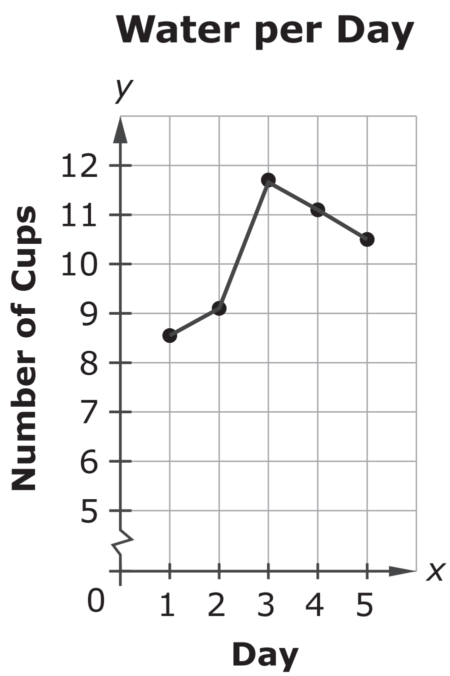
What number is one-hundredth more than 732.12?

Write your response in the shaded box below.



Some items may have more than one box, so read each item carefully. Your answers for the items with response boxes may contain whole numbers, fractions, or decimals.

1. The graph shows the number of cups of water Beatrice drinks each day for five days.



How many cups of water does Beatrice drink on Day 4?

Ⓐ 11.8

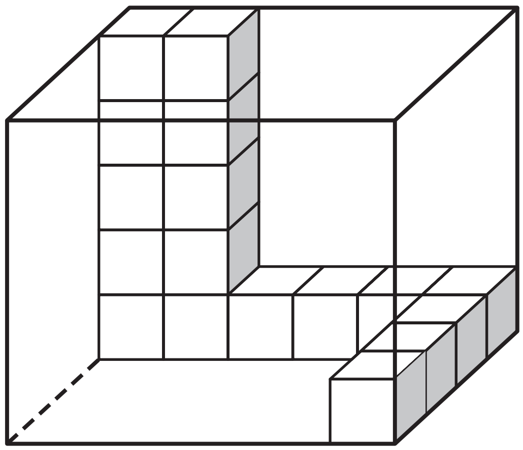
Ⓑ 11.2

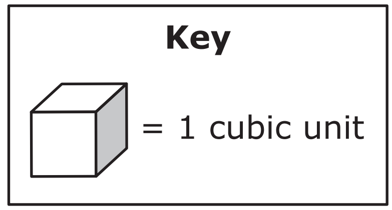
Ⓒ 10.5

Ⓓ 10.2

Go On

2. A right rectangular prism is partially filled with unit cubes, as shown.





How many more unit cubes are needed to completely fill the right rectangular prism?

Write your response in the shaded box below.

3. Fill in bubbles to match each expression with the statement that describes it.

	The sum of 4 and 3, multiplied by 7	**The product of 3 and 7, added to 4**	**The product of 4 and 7, added to 3**
3 + 7 × 4	Ⓐ	Ⓑ	Ⓒ
4 + 3 × 7	Ⓓ	Ⓔ	Ⓕ
(4 + 3) × 7	Ⓖ	Ⓗ	Ⓘ

Go On

4. This question has **two** parts.

Jerry incorrectly subtracts 3.6 from 3.966, as shown.

$$\begin{array}{r} 3.9\,6\,6 \\ -\quad 3.6 \\ \hline 3.9\,3\,0 \end{array}$$

Part A

Which statement **best** describes Jerry's mistake?

Ⓐ Jerry should have regrouped.
Ⓑ Jerry should have subtracted 3.966 from 3.6.
Ⓒ The decimal is in the wrong place in Jerry's answer.
Ⓓ When setting up the problem, Jerry did not line up the place values.

Part B

What is the correct answer?

Ⓐ 0.3
Ⓑ 0.366
Ⓒ 3.606
Ⓓ 393.0

5. A number is shown.

35.714

Select words to show one way to decompose the number. For each box, fill in the bubble before the word that is correct.

35 [Ⓐ ones / Ⓑ tens] + 71 [Ⓐ tenths / Ⓑ hundredths / Ⓒ thousandths] + 4 [Ⓐ tenths / Ⓑ hundredths / Ⓒ thousandths]

6. An expression with a missing number is shown.

$$7 + 3 \times 2 \times \square$$

Select all the expressions that are equivalent to the given expression.

- Ⓐ $(7 + 3) \times 2 \times \square$
- Ⓑ $7 + (3 \times 2) \times \square$
- Ⓒ $7 + 3 \times (2 \times \square)$
- Ⓓ $7 + (3 \times 2 \times \square)$
- Ⓔ $(7 + 3) \times (2 \times \square)$

7. The table shows the amount of rainwater three students gather.

Student	**Amount**
Damien	$1\frac{1}{8}$ gallons
Taylor	1 gallon and 4 cups
Veronica	$19\frac{1}{2}$ cups

Who gathers the most rainwater, and who gathers the least rainwater? For each blank, fill in the bubble **before** the name that is correct.

Most: ___________ [Ⓐ Damien Ⓑ Taylor Ⓒ Veronica]

Least: ___________ [Ⓐ Damien Ⓑ Taylor Ⓒ Veronica]

8. Rebekah makes a quilt using red, blue, and purple fabric.

- She uses $3\frac{1}{2}$ total yards of fabric.
- She uses exactly $2\frac{3}{5}$ yards of purple fabric.

What are possible lengths, in yards, of red and blue fabric that Rebekah could use?

Write your responses in the shaded boxes below.

Red:

Blue:

9. An expression is shown.

$$12 \times \frac{5}{6}$$

Complete the sentence about the expression. For each box, fill in the bubble before the phrase that is correct.

The value of the expression is predicted to be

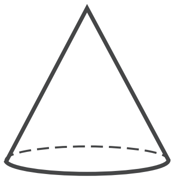
- Ⓐ equal to 12
- Ⓑ less than 12
- Ⓒ greater than 12

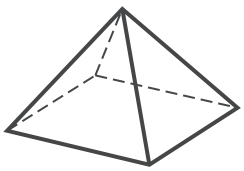
because 12 is being multiplied by a fraction

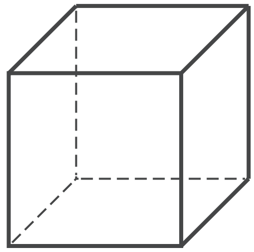
- Ⓐ equal to 1
- Ⓑ less than 1
- Ⓒ greater than 1

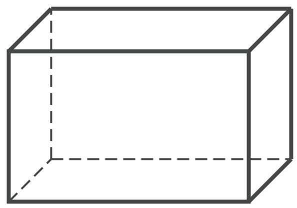
.

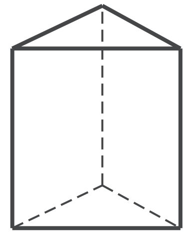
10. Select all the figures that have parallel bases.

A 

B 

C 

D 

E 

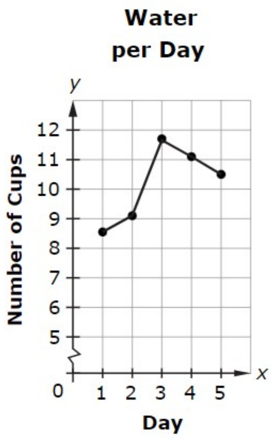
Grade 5

FAST Mathematics

Sample Items
Answer Key

The Grade 5 FAST Mathematics Sample Items Answer Key provides the correct response(s) for each item on the sample test. The sample items and answers are not intended to demonstrate the length of the actual test, nor should student responses be used as an indicator of student performance on the actual test.

1. The graph shows the number of cups of water Beatrice drinks each day for five days.



How many cups of water does Beatrice drink on Day 4?

Ⓐ 11.8

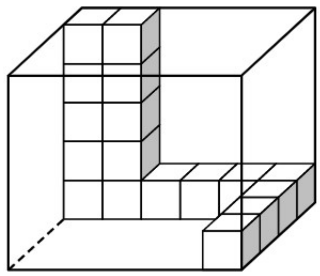
● 11.2

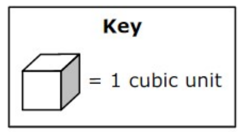
Ⓒ 10.5

Ⓓ 10.2

<u>*Option B*</u>*:* **This answer is correct.** *The student identified that the height of the data point is between 11 and 11.5, so 11.2 is a reasonable estimation.*

2. A right rectangular prism is partially filled with unit cubes, as shown.





How many more unit cubes are needed to completely fill the right rectangular prism?

Other correct responses: *any equivalent value*

3. Match each expression with the statement that describes it.

	The sum of 4 and 3, multiplied by 7	The product of 3 and 7, added to 4	The product of 4 and 7, added to 3
$3 + 7 \times 4$	☐	☐	☑
$4 + 3 \times 7$	☐	☑	☐
$(4 + 3) \times 7$	☑	☐	☐

4. This question has **two** parts.

Jerry incorrectly subtracts 3.6 from 3.966, as shown.

$$\begin{array}{r} 3.9\,6\,6 \\ -\quad\;\; 3.6 \\ \hline 3.9\,3\,0 \end{array}$$

Part A
Which statement **best** describes Jerry's mistake?

- Ⓐ Jerry should have regrouped.
- Ⓑ Jerry should have subtracted 3.966 from 3.6.
- Ⓒ The decimal is in the wrong place in Jerry's answer.
- ● When setting up the problem, Jerry did not line up the place values.

Part B
What is the correct answer?

- Ⓐ 0.3
- ● 0.366
- Ⓒ 3.606
- Ⓓ 393.0

Part A

Option D: **This answer is correct.** *The student recognized that the subtraction problem was set up incorrectly because the decimals (and therefore place values) are not in line.*

Part B

Option B: **This answer is correct.** *The student correctly subtracted 3.6 from 3.966.*

5. A number is shown.

35.714

Select words to show one way to decompose the number.

35 [ones] + 71 [hundredths] + 4 [thousandths]

Go On

6. An expression with a missing number is shown.

$7 + 3 \times 2 \times \square$

Select all the expressions that are equivalent to the given expression.

- [ ] $(7 + 3) \times 2 \times \square$
- [x] $7 + (3 \times 2) \times \square$
- [x] $7 + 3 \times (2 \times \square)$
- [x] $7 + (3 \times 2 \times \square)$
- [ ] $(7 + 3) \times (2 \times \square)$

Option B: **This answer is correct.** *The student correctly identified that this expression is equivalent to the given expression.*

Option C: **This answer is correct.** *The student correctly identified that this expression is equivalent to the given expression.*

Option D: **This answer is correct.** *The student correctly identified that this expression is equivalent to the given expression.*

7. The table shows the amount of rainwater three students gather.

Student	Amount
Damien	$1\frac{1}{8}$ gallons
Taylor	1 gallon and 4 cups
Veronica	$19\frac{1}{2}$ cups

Who gathers the most rainwater, and who gathers the least rainwater?

Most: Taylor

Least: Damien

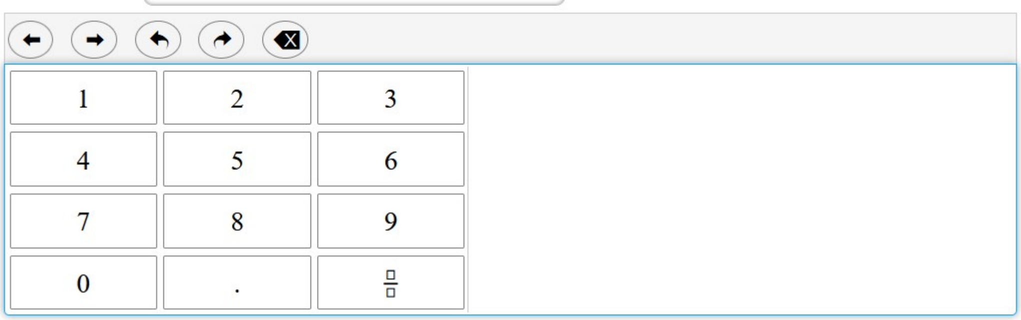
8. Rebekah makes a quilt using red, blue, and purple fabric.

- She uses $3\frac{1}{2}$ total yards of fabric.
- She uses exactly $2\frac{3}{5}$ yards of purple fabric.

What are possible lengths, in yards, of red and blue fabric that Rebekah could use?

Red: $\frac{3}{10}$

Blue: $\frac{6}{10}$



Other correct responses: *any two nonzero values that sum to* $\frac{9}{10}$

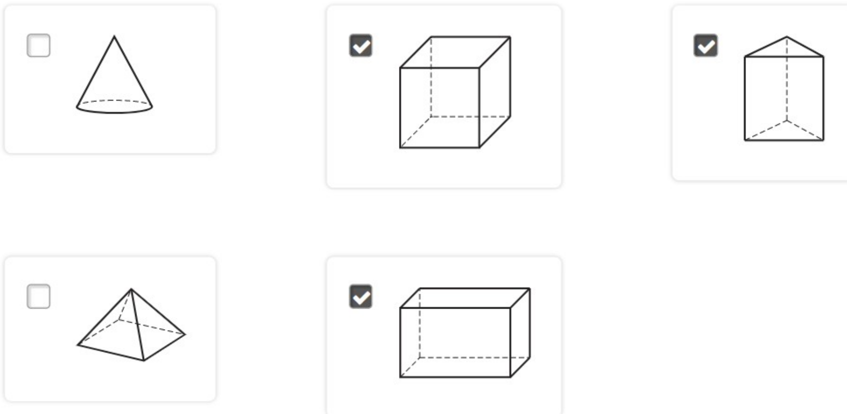
9. An expression is shown.

$12 \times \frac{5}{6}$

Complete the sentence about the expression.

The value of the expression is predicted to be [less than 12] because 12 is being multiplied by a fraction [less than 1].

10. Select all the figures that have parallel bases.



<u>Option C</u>: **This answer is correct.** *The student identified a figure with parallel bases.*

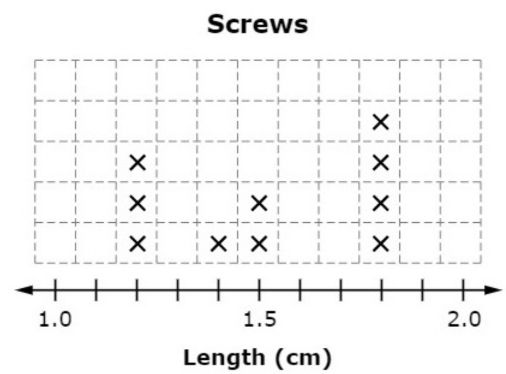
<u>Option D</u>: **This answer is correct.** *The student identified a figure with parallel bases.*

<u>Option E</u>: **This answer is correct.** *The student identified a figure with parallel bases.*

11. The lengths, in centimeters (cm), of 10 screws are shown in the table.

Number of Screws	Length of Screws (cm)
1	1.4
4	1.8
3	1.2
2	1.5

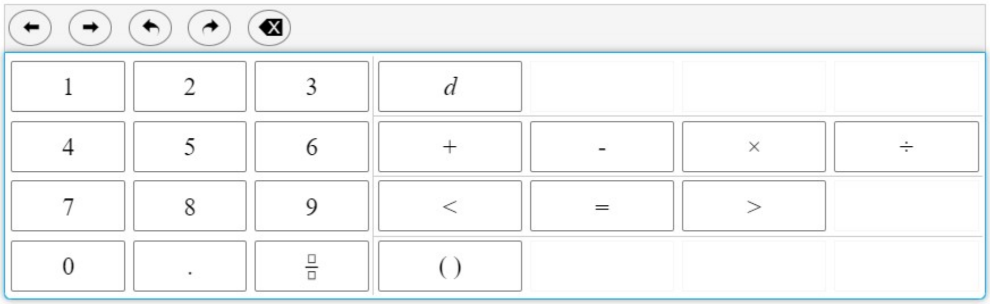
Click above the number line to create a line plot showing the lengths of the screws.



12. There is a total of 85 chocolate doughnuts and 77 strawberry doughnuts in 18 boxes. Each box has d doughnuts.

Create an equation using d that represents the number of doughnuts in each box.

$$(85+77)\div d=18$$



Other correct responses: *any equivalent equation*

Go On

13. Jane uses the expression $3x + 6$ to create the input-output table shown, where x represents a whole number.

Input (x)	Output
0	6
1	9
2	11
3	15

She makes an error in the output column.

Complete the sentence to describe Jane's error.

The output when x is 2 should be 12.

Other correct responses: *any equivalent value*

14. This question has **two** parts.

An expression is shown.

Part A
Click on the operation that is performed last when evaluating the expression.

$8 - (9 + 2) \div 11$

Part B
What is the value of the expression?

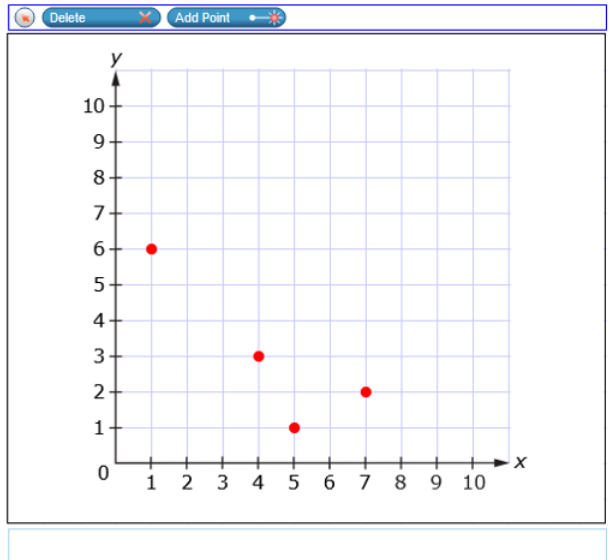
7

Other correct responses: *for Part B, any equivalent value*

15. A table of ordered pairs is shown.

x	*y*
1	6
7	2
4	3
5	1

Use the Add Point tool to plot the points represented in the table.



16.

Jonah has 3 limes that weigh a total of 8 ounces.

Click on the grid to plot a point on the coordinate plane to represent Jonah's limes.